

Program 1 : Build a docker container from a custom Docker file

File structure

index.js

package.json

Dockerfile

Run the following commands to build and run the image:

1. To build the image:

```
$ docker build -t program1 .
```

2. To view created images:

```
$ docker images
```

3. To run the container:

```
$ docker run -d -p 4000:4000 --name program1 program1
```

4. To see running containers:

```
$ docker ps
```

5. To test the application:

```
$ curl http://localhost:4000
```

Execution Output:

Nov 9 16:57

girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab1

girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab1\$ ls

Dockerfile index.js node_modules package.json package-lock.json

girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab1\$ docker build -t program1 .

[+] Building 46.8s (10/10) FINISHED

==> [internal] load build definition from Dockerfile

==> ==> transferring dockerfile: 168B

==> [internal] load metadata for docker.io/library/node:18-alpine

==> [internal] load .dockerignore

==> ==> transferring context: 2B

==> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca89d9e

==> ==> resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca89d9e

==> ==> sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca89d9e 7.67kB / 7.67kB

==> ==> sha256:929b04d7c782f04f615cf785488fed452b6569f87c73ff66ad553a7554f0006 1.72kB / 1.72kB

==> ==> sha256:ee7c6cd7c1886ecc802ad6cedf3a8ec1ea27d1fb96162bf03dd3710839b0da 6.18kB / 6.18kB

==> ==> sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB

==> ==> sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feaf161b2b5a3d0e 40.01MB / 40.01MB

==> ==> sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 1.26MB / 1.26MB

==> ==> extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870

==> ==> sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B

==> ==> extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feaf161b2b5a3d0e

==> ==> extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3

==> ==> extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549

==> [internal] load build context

==> ==> transferring context: 28.57kB

==> [2/5] WORKDIR /app

==> [3/5] COPY package*.json ./

==> [4/5] RUN npm install

==> [5/5] COPY index.js .

==> ==> exporting to image

==> ==> exporting layers

==> ==> writing image sha256:08c48af5c92ec75bc2a049c81138d31bb9d4f551276e9bcdad1d5fb39e19a979

==> ==> naming to docker.io/library/program1

girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab1\$ docker run -d -p 4000:4000 program1

38c67a115ddf788ea541ca571e11cafc4517141df3b937cf79ccbc1ad8f042

girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab1\$ curl http://localhost:4000

Hello wassup dockergirish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab1\$